

Ethan Huang

ethan.huang299@gmail.com | 613-606-3023 | ethan-huang-profile.com | github.com/EthanHuanggg

Education

- Carleton University, **Bachelor of Electrical Engineering** Sept. 2024 - April 2028
- GPA: **3.9 / 4.0**
- Accolades: OCDSB Honour Society Award, Entrance Scholarship, Golden Key Award (*Top 15% of program*)
- Coursework: Electronics 1, Switching Circuits, Circuits & Signals, Object-Oriented Software Development

Technical Skills

Languages: Python, C, Java, Linux, HTML, CSS, Verilog.

Technologies: KiCAD, Fusion 360, Arduino, Bambu Studio, Xilinx Vivado, MATLAB/Matplotlib, Excel, Git, Multisim Live, FPGA Design, Oscilloscopes, Multimeter.

Projects

FM Stereo Radio Circuit (In Progress) - March 2026

- Designed and fabricated custom Arduino Nano-based FM stereo radio **PCB** in **KiCAD**
- Integrated RDA5807M FM stereo tuner operating at 87-108 MHz with full **I2C** control
- Implemented 5V-to-3.3V bidirectional I2C level shifting with dual BSS138 MOSFETs

Arduino Robot Arm - Dec. 2025

- Modeled and fabricated structural components in **Fusion 360 3D AutoCAD** software using FDM **3D printing**, performing mechanical assembly, wiring, and system debugging.
- Designed Arduino-based robotic arm control system integrating servo motors, potentiometer feedback, and microcontroller control (**C++ / Arduino IDE**).
- Implemented real-time position control and multi-axis actuation testing.

Real-Time Hand Tracking Software - Jan. 2026

- Developed a real-time hand tracking system using **Python**, OpenCV, and MediaPipe.
- Implemented image acquisition, RGB color space conversion, landmark detection, and numerical feature extraction.
- Applied signal filtering to reduce noise and logged time-series data for analysis and visualization using NumPy and Matplotlib.

Work Experience & Extra Curriculars

Ciena x SCESoc Coding Competition @Carleton University (March 2026)

- Debugged and extended **WaveServer MINI OTN** router simulator, enhancing Ciena's provided high-level design with packet processing and port management features
- Implemented **UDP** socket interface, stateful connection tracking, and resource allocation logic for multi-client optical transponder simulation under production-like constraints
- Analyzed existing codebase to identify and resolve bugs in **C**, implement new functionality, and ensure protocol compliance for optical transponder simulation

Server Assistant @ The Marshes Golf Club, Kanata, ON (April 2026 - present)

- Supported front-of-house operations at Ironstone Grill by assisting servers with food delivery, table setup, and clearing
- Maintained clean and organized dining areas to ensure efficient service flow and positive guest experience
- Collaborated with kitchen and service staff in a fast-paced environment to meet high service standards
- Provided attentive customer support by responding to guest needs promptly and professionally